ANALYSIS I — SHEET 1 (for Week 3 tutorials)
Real Numbers, Arithmetic, Order, (Real) Modulus.

1. Let $a, b, c \in \mathbb{R}$. Prove carefully, *from the given Axioms for the Real Numbers, or using anything proven in lectures,* that:

(i) if $ab = ac$ and $a \neq 0$, then $b = c$;

(ii) if $ab = 0$, then $a = 0$ or $b = 0$;

(iii) $-(a + b) = (-a) + (-b)$;

(iv) $a \geqslant b$ if and only if $a + c \geqslant b + c$;

(v) if $a < b$ and $c < 0$, then $ac > bc$.

2. Let $a \in \mathbb{R} \setminus \{0\}$. Show that $a^{k+1} = a^k a$ for $k \in \mathbb{Z}$. Deduce the law of indices:

$$a^m a^n = a^{m+n} \qquad \text{for } m, n \in \mathbb{Z}.$$

3. (i) Let $a, b \in \mathbb{R}$. Prove that

$$|a + b| = |a| + |b| \text{ if and only if } (a \geqslant 0 \text{ and } b \geqslant 0) \text{ or } (a \leqslant 0 \text{ and } b \leqslant 0).$$

(ii) Carefully define $\max(a, b)$, the maximum of two real numbers a, b, saying which axiom(s) show that your definition is well-defined. Prove that

$$\max(a, b) = \frac{1}{2}(a + b) + \frac{1}{2}|a - b|.$$

4. Carefully determine all $x \in \mathbb{R}$ such that

$$4 < |x + 2| + |x - 1| < 5.$$

5. For each of the following conditions, find and sketch the set of all pairs $(x, y) \in \mathbb{R}^2$ satisfying the condition:

$$\text{(i)} \quad |x| + |y| = 1; \qquad \text{(ii)} \quad |x - y| + |x + y| = 1; \qquad \text{(iii)} \quad \max(|x|, |y|) = 1.$$

6. For $n = 1, 2, 3, \ldots$ let

$$a_n = \left(1 + \frac{1}{n}\right)^n \text{ and } b_n = \left(1 + \frac{1}{n}\right)^{n+1}.$$

(i) Show that the inequality $a_n \leqslant a_{n+1}$ can be rearranged as

$$\left(\frac{n(n+2)}{(n+1)^2}\right)^{n+1} \geqslant \frac{n}{n+1}. \tag{1}$$

(ii) By applying Bernoulli's inequality to the left hand side verify inequality (1).

(iii) In a similar manner, show $b_{n+1} \leqslant b_n$ for all n.

(iv) Note $a_n \leqslant a_{n+1} \leqslant b_{n+1} \leqslant b_n$. Show that $a_m \leqslant b_n$ for all m, n and deduce that $a_n < 3$ for all n.

Extension Exercises (Optional)

7. Assuming the existence of $\sqrt{2}$ and that $\left(\sqrt{2}\right)^2 = 2$, show that

$$\mathbb{Q}\left[\sqrt{2}\right] = \left\{q_1 + q_2\sqrt{2} \mid q_1, q_2 \in \mathbb{Q}\right\}$$

is a field. [*Hint: many of the field axioms are automatically inherited from the real numbers as* $\mathbb{Q}\left[\sqrt{2}\right] \subseteq \mathbb{R}$.]

8. For $n \geqslant 2$, let $\mathbb{Z}_n$ denote the integers modulo n.
(i) Find all the roots of $x^2 - 1$ in $\mathbb{Z}_8$.
(ii) Find two distinct ways to factorize $x^2 - 1$ over $\mathbb{Z}_8$.
(iii) Which of the field axioms does $\mathbb{Z}_8$ satisfy?
(iv) Show that all the field axioms are satisfied by $\mathbb{Z}_7$.
(v) Is there a subset $\mathbb{P}$ of $\mathbb{Z}_7$ which satisfies the order axioms?

9. There is a field $\mathbb{F}_4$, with four elements. Two of the elements are necessarily 0 and 1 and we will label the other two elements x and y.
(i) From the field axioms, explain why $xy = 1$ and complete the multiplication table of $\mathbb{F}_4$.
(ii) Show that $1 + 1 = x$ leads to a contradiction and deduce that $1 + 1 = 0$.
(iii) Deduce further that $1 + x = y$.
(iv) Complete the addition table for $\mathbb{F}_4$.
(v) Show that the four matrices

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

with entries mod 2, form an instance of $\mathbb{F}_4$.

10. Let $\leqslant$ be a total order which also satisfies Propositions 1.28 and 1.29. Show that the set

$$P = \{x \in \mathbb{R} \mid 0 \leqslant x\} \setminus \{0\}$$

satisfies the axioms P1, P2, P3 in the definition of $\mathbb{P}$.

11. (i) Find c as determined by Proposition 1.46 such that $2^n \geqslant cn^3$ for all natural numbers n.
(ii) What is the maximal value c such that $2^n \geqslant cn^3$ for all natural numbers n?